

Particle Physics 2: Standard Model
Lecture 7: Broken Symmetry, Goldstone Bosons, and Gauge
Invariance

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Chapter 1

Broken Symmetry, Goldstone Bosons, and Gauge Invariance

We want to understand how a photon-like particle can acquire a mass. That is the destination, but it is not a sensible starting point. A direct mass term for a gauge field appears to destroy the very gauge invariance that makes the theory consistent. Before we can see the escape route, we need two ideas in working form:

1. spontaneous symmetry breaking, especially for a continuous symmetry;
2. gauge invariance, in which the symmetry operation may vary from one spacetime point to another.

Neither idea can be replaced by a slogan. We shall build the first from a magnet, translate it into scalar field theory, and discover the massless Goldstone mode. We shall then take the same $U(1)$ symmetry and make it local, which forces us to introduce a vector potential and a covariant derivative. Only at the end will the two threads be placed side by side. Their actual union—the Higgs phenomenon—is the subject of the next lecture.

1.1 A Symmetric Law with an Asymmetric Ground State

1.1.1 The two-state magnet

Begin with a lattice of small magnets. Each spin can point up or down, and neighboring spins lower their energy by pointing in the same direction. On a three-dimensional cubic lattice a spin has six nearest neighbors, but the precise coordination number is not important. What matters is the rule

$$E_{\parallel} < E_{\text{antiparallel}}. \quad (1.1)$$

The microscopic rule is unchanged if every spin is reversed:

$$\uparrow \longleftrightarrow \downarrow. \quad (1.2)$$

Nevertheless, the lowest-energy configurations are not individually symmetric. They are

$$\{\uparrow\uparrow\uparrow \cdots\}, \quad \{\downarrow\downarrow\downarrow \cdots\}. \quad (1.3)$$

The equations allow both, but a cold macroscopic sample must choose one.

This choice is extraordinarily sensitive. Fix one spin very far away to point up. Its neighbors prefer to follow it, their neighbors prefer to follow them, and the information propagates through the sample. A perturbation that is arbitrarily small compared with the total system can select the orientation of the whole ground state. That is the instability meant by *spontaneous* symmetry breaking.

Question from the audience. Can a neutral particle orient itself in a magnetic field? ^a

Response. Yes. Electric neutrality does not imply the absence of a magnetic moment. The neutron is neutral but has a magnetic moment because of its internal charged constituents. A magnetic field can therefore distinguish its spin orientations.

^aLecture timestamp: 00:05:14.

Question from the audience. If the system changes orientation, does the change occur everywhere at once? Can several reorientation waves coexist? ^a

Response. The change propagates. One region tips its neighbors, rather like a three-dimensional array of dominoes, and the disturbance travels as a wave. Different regions can begin choosing independently, so several fronts and several domains may be present before one orientation wins. We shall return to precisely these waves when the symmetry becomes continuous.

^aLecture timestamp: 00:05:41–00:06:38.

1.1.2 Explicit breaking is a different theory

Now place the sample in an external magnetic field pointing upward. An isolated spin already has lower energy when it points up. The transformation $\uparrow \leftrightarrow \downarrow$ no longer leaves the energy unchanged: the preference has been inserted into the law itself. This is *explicit* symmetry breaking.

The difference can be tested without knowing how the sample was prepared. Take a very large specimen, hold its left boundary up, and hold its right boundary down. If a bulk magnetic field explicitly favors up, the right-hand constraint penetrates only a short distance. A macroscopic down region would pay an energy proportional to its volume, so the bulk cannot remain divided into two equally good phases.

Without the external bias, up and down have the same bulk energy. The lowest-energy constrained state then contains an up domain, a down domain, and a finite transition layer between them. The interface need not sit at the geometrical center, but there is no volume pressure forcing one phase to erase the other. This interface is a *domain wall*, the characteristic defect of this broken discrete symmetry.

Question from the audience. Does the argument require interactions only between nearest neighbors? ^a

Response. No. A second-neighbor interaction that also favors alignment changes details but not the conclusion. More interesting behavior appears if different ranges compete—nearest neighbors prefer alignment while next-nearest neighbors prefer anti-alignment, for example. That produces frustration and more complicated ordered phases, but it does not alter the basic distinction being made here.

^aLecture timestamp: 00:18:18.

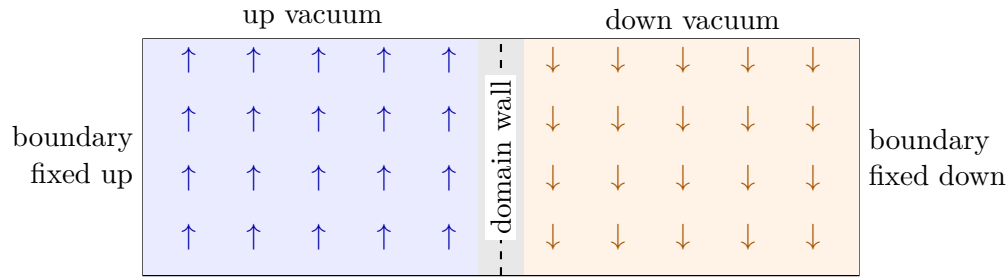


Figure 1.1: The boundary test for spontaneous breaking of a discrete symmetry. Both bulk orientations have the same energy, so a wall can separate macroscopic domains.

Question from the audience. Does *spontaneous* refer to the large derivative inside the wall?^a

Response. No. It refers to the fact that the underlying energy has no up–down preference while the selected ground state does. The dramatic response to an arbitrarily small perturbation reveals an instability. The gradient merely tells us how much the interface costs.

^aLecture timestamp: 00:19:05.

Question from the audience. Was the material already in one of the two states, or was its state initially unknown?^a

Response. That depends on its history. At high temperature thermal motion can leave up and down spins mixed with nearly zero average magnetization. As the material cools, patches choose different orientations, domain walls form, and the domains coarsen. Chance and tiny environmental perturbations determine which vacuum finally occupies most of the sample.

^aLecture timestamp: 00:19:40.

1.2 The Same Argument as Scalar Field Theory

Replace the spins by a real scalar field $\phi(\mathbf{x}, t)$. With metric and normalization conventions fixed once and for all, write

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi), \quad \mathcal{E} = \frac{1}{2} \dot{\phi}^2 + \frac{1}{2} (\nabla \phi)^2 + V(\phi). \quad (1.4)$$

The relativistic shorthand in $\mathcal{L}$ contains the positive time-derivative term and negative spatial-gradient term; the Hamiltonian energy density has both contributions positive.

Choose an even potential with two degenerate minima,

$$V(\phi) = \frac{\lambda}{4} (\phi^2 - f^2)^2, \quad \lambda > 0. \quad (1.5)$$

The Lagrangian is invariant under $\phi \mapsto -\phi$, while either uniform vacuum

$$\phi(\mathbf{x}) = +f, \quad \text{or} \quad \phi(\mathbf{x}) = -f \quad (1.6)$$

chooses one side. A potential with a unique minimum at $\phi = 0$, by contrast, has a symmetric vacuum and is not spontaneously broken.

Fix $\phi = +f$ on one side of a sample and $\phi = -f$ on the other. The field cannot jump without cost, because

$$E_{\text{wall}} = \int dx \left[\frac{1}{2} \left(\frac{d\phi}{dx} \right)^2 + V(\phi) - V(f) \right]. \quad (1.7)$$

An abrupt jump makes $d\phi/dx$ enormous. The minimum-energy wall therefore spreads the change over a finite width, balancing gradient energy against the potential cost of passing between the minima. The familiar kink solution for the potential in Eq. (1.5) is

$$\phi_{\text{wall}}(x) = f \tanh \left[f \sqrt{\frac{\lambda}{2}} (x - x_0) \right], \quad (1.8)$$

an explicit realization of the smooth transition described at the board.¹

Tilt the double well slightly and one minimum becomes lower. The Lagrangian itself is then asymmetric. A wall can still appear temporarily, but the higher-energy or false vacuum exerts a pressure on it; the wall moves and the favored phase expands. This is the field-theory version of explicit breaking by an external magnetic field.

So far the symmetry has been discrete. Particle physics is dominated by continuous groups such as $U(1)$, $SU(2)$, and $SU(3)$. A continuous symmetry changes the defect story and introduces a new kind of particle.

1.3 A Complex Field and Global $U(1)$

1.3.1 Rotations in an internal plane

Let

$$\phi = \phi_R + i\phi_I, \quad \phi^* = \phi_R - i\phi_I, \quad \phi^*\phi = \phi_R^2 + \phi_I^2. \quad (1.9)$$

The ϕ_R and ϕ_I axes form an internal field space, not two directions in ordinary space. A complex scalar is simply two real scalar fields packaged so that rotations in this internal plane are manifest.

Use the Lagrangian

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - V(\phi^* \phi). \quad (1.10)$$

For a complex field the conventional factor $1/2$ is absent, because this single product already contains the kinetic terms of both real components. The transformation

$$\phi'(x) = e^{i\theta} \phi(x), \quad \phi^{*'}(x) = e^{-i\theta} \phi^*(x) \quad (1.11)$$

rotates every field value through the same angle. Here θ is constant in space and time. The prime labels the transformed field; it is not a derivative.

Because θ is constant,

$$\partial_\mu \phi' = e^{i\theta} \partial_\mu \phi, \quad \partial_\mu \phi^{*'} = e^{-i\theta} \partial_\mu \phi^*. \quad (1.12)$$

The phases cancel in the kinetic term, and $\phi^{*'}\phi' = \phi^*\phi$. Thus every potential that depends only on $\phi^*\phi$ has the same symmetry. A term depending on ϕ_R alone would pick a direction in the internal plane and break it explicitly.

¹Editorial clarification: The lecture argues from the shape of the potential and the gradient term but does not solve the field equation. Equation (1.8) is included as the standard solution of that stated model, making the qualitative wall concrete without changing the argument.

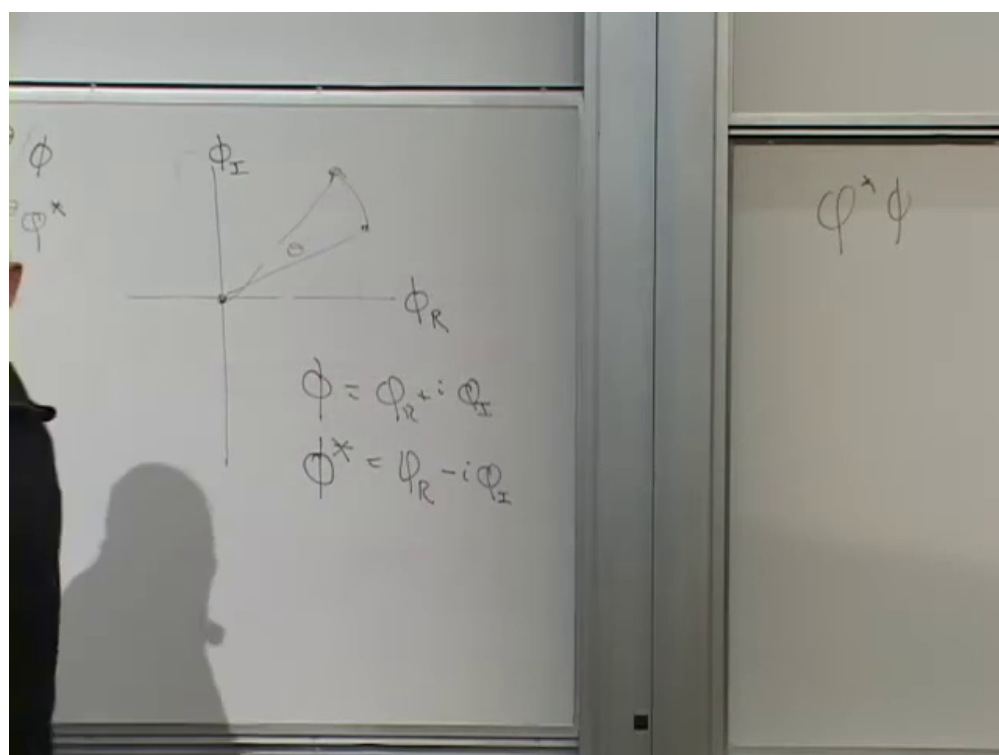


Figure 1.2: The complex field on the internal (ϕ_R, ϕ_I) -plane. The board records the decomposition of ϕ and ϕ^* and the invariant radius $\phi^* \phi$.

Lecture frame: 00:33:33.

1.3.2 A point minimum and a circle of minima

There are two qualitatively different symmetric potentials. The first has its only minimum at the origin, for example

$$V_{\text{unbroken}} = m^2 \phi^* \phi, \quad m^2 > 0. \quad (1.13)$$

Its graph is a rotationally symmetric paraboloid. The vacuum $\phi = 0$ stays fixed under every rotation, so the symmetry is unbroken.

Now choose

$$V_{\text{broken}} = -a \phi^* \phi + b(\phi^* \phi)^2, \quad a, b > 0. \quad (1.14)$$

The negative quadratic term drives the field away from the origin, while the positive quartic term prevents it from running to infinity. Writing $r^2 = \phi^* \phi$, minimization gives

$$\frac{dV}{d(r^2)} = -a + 2br^2 = 0, \quad r^2 = f^2 = \frac{a}{2b}. \quad (1.15)$$

The minima therefore form a circle. The potential is perfectly rotationally symmetric, yet a uniform ground state must choose one point on that circle.

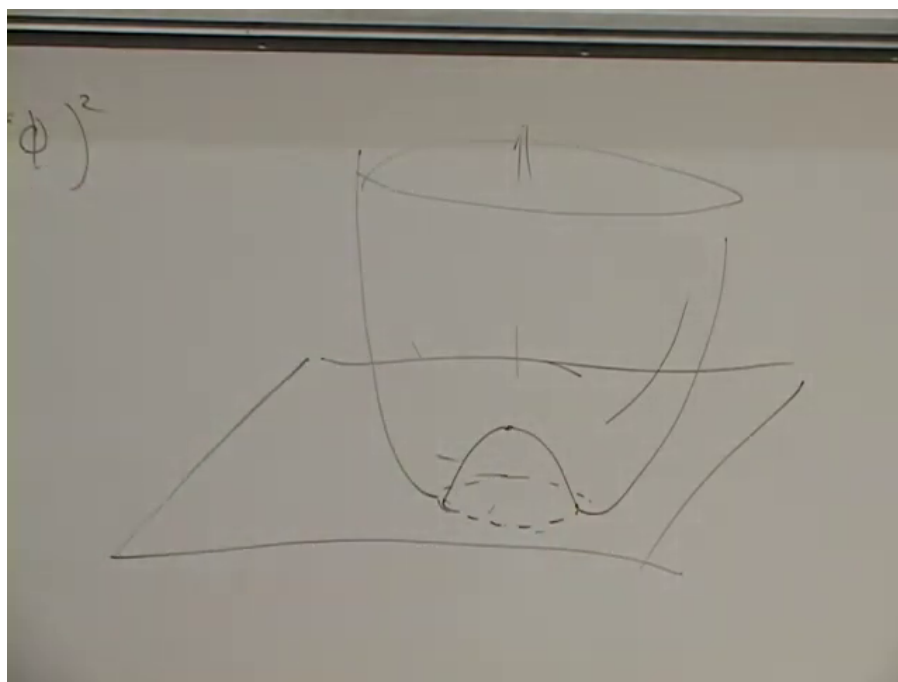


Figure 1.3: The symmetry-breaking potential drawn as a surface of revolution. The origin is unstable and the degenerate minima form the circular trough.

Lecture frame: 00:37:25.

Think of the field as a little arrow in this internal plane at every point of ordinary space. The potential asks every arrow to have length f . The gradient energy asks nearby arrows to point in nearly the same direction. The ground state consequently has the same angle everywhere, but no angle is preferred by the law. That is continuous spontaneous symmetry breaking.

1.4 From a Domain Wall to a Smooth Twist

Repeat the boundary experiment. At the left boundary fix the field at a vacuum point A ; at the right fix it at another vacuum point B . A localized jump would make the gradient large. Unlike the discrete case, however, the field can remain at the bottom of the potential while changing continuously around the circle.

At low energy write the field as $\phi = f e^{i\alpha(x)}$. In one spatial dimension the relevant energy is

$$E[\alpha] = f^2 \int_0^L dx \left(\frac{d\alpha}{dx} \right)^2. \quad (1.16)$$

For fixed endpoint angles, variation gives

$$\frac{d^2\alpha}{dx^2} = 0, \quad \alpha(x) = \alpha_A + \frac{\alpha_B - \alpha_A}{L} x. \quad (1.17)$$

The preferred configuration spreads the twist uniformly through the sample. Its energy scales as

$$E_{\text{twist}} = f^2 \frac{(\alpha_B - \alpha_A)^2}{L}, \quad (1.18)$$

so it can be made arbitrarily cheap by increasing L .

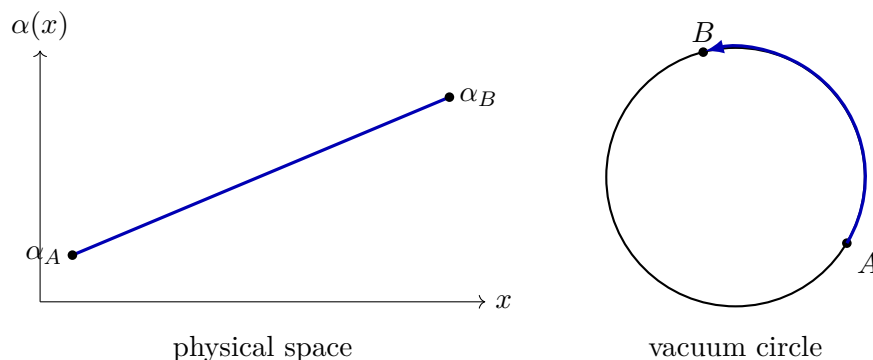


Figure 1.4: The same low-energy configuration in two descriptions. The angle varies linearly across physical space while the field follows the short arc of the vacuum circle in internal space.

Question from the audience. There can be two ways around the circle. Which path does the field take, and can it become trapped on the longer one? ^a

Response. For generic endpoints the shorter angular route has the smaller gradient and therefore the lower energy. Diametrically opposite endpoints make the two simplest routes degenerate. A configuration prepared by slowly moving an endpoint may retain an extra winding and become trapped behind an energy barrier, but for a specified pair of ordinary endpoints the untwisted shortest route is the ground state.

^aLecture timestamp: 00:41:54.

Question from the audience. Could a path leave the bottom of the trough and cut across the interior, becoming shorter overall?^a

Response. In the low-energy, long-sample regime it does not pay to climb the radial potential. Keeping the radius at its vacuum value and varying the angle linearly minimizes the effective energy, as the calculus-of-variations calculation above shows. For a finite system with a specific potential one can minimize radial and angular motion together, but the Goldstone conclusion concerns the arbitrarily long-wavelength limit in which the smooth motion along the trough wins.

^aLecture timestamp: 00:43:32.

Question from the audience. What changes if the circular trough is tilted so that one angle is lower?^a

Response. Then the symmetry is explicitly broken. One point becomes the true ground state, motion along the circle acquires a restoring force, and the angular excitation is no longer exactly massless.

^aLecture timestamp: 00:44:33.

1.5 Mass as the Cost of a Zero-Momentum Oscillation

The smooth twist reveals a field whose energy comes only from gradients. To understand why that means a massless particle, compare the relativistic dispersion relation

$$\omega^2 = \mathbf{k}^2 + m^2. \quad (1.19)$$

A wave $e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}$ carries momentum $\mathbf{k}$. If $m = 0$, its energy tends to zero as the wavelength tends to infinity: $\omega = |\mathbf{k}| \rightarrow 0$. If $m \neq 0$, a spatially homogeneous mode $\mathbf{k} = 0$ still oscillates with frequency $\omega = m$. The mass is the irreducible zero-momentum energy.

This same distinction is visible in the Lagrangian. A term

$$-\frac{1}{2}m^2\varphi^2 \quad (1.20)$$

supplies a restoring force even for a homogeneous displacement. A field with only derivative terms has no such restoring force: make its wavelength longer, and its gradient energy approaches zero.

1.5.1 Polar variables separate the two modes

Write the broken complex field as

$$\phi(x) = \rho(x)e^{i\alpha(x)}. \quad (1.21)$$

Both ρ and α are fields. Differentiation gives

$$\partial_\mu\phi = e^{i\alpha}(\partial_\mu\rho + i\rho\partial_\mu\alpha), \quad (1.22)$$

$$\partial_\mu\phi^* = e^{-i\alpha}(\partial_\mu\rho - i\rho\partial_\mu\alpha). \quad (1.23)$$

The cross terms cancel, leaving

$$\partial_\mu\phi^*\partial^\mu\phi = (\partial_\mu\rho)(\partial^\mu\rho) + \rho^2(\partial_\mu\alpha)(\partial^\mu\alpha). \quad (1.24)$$

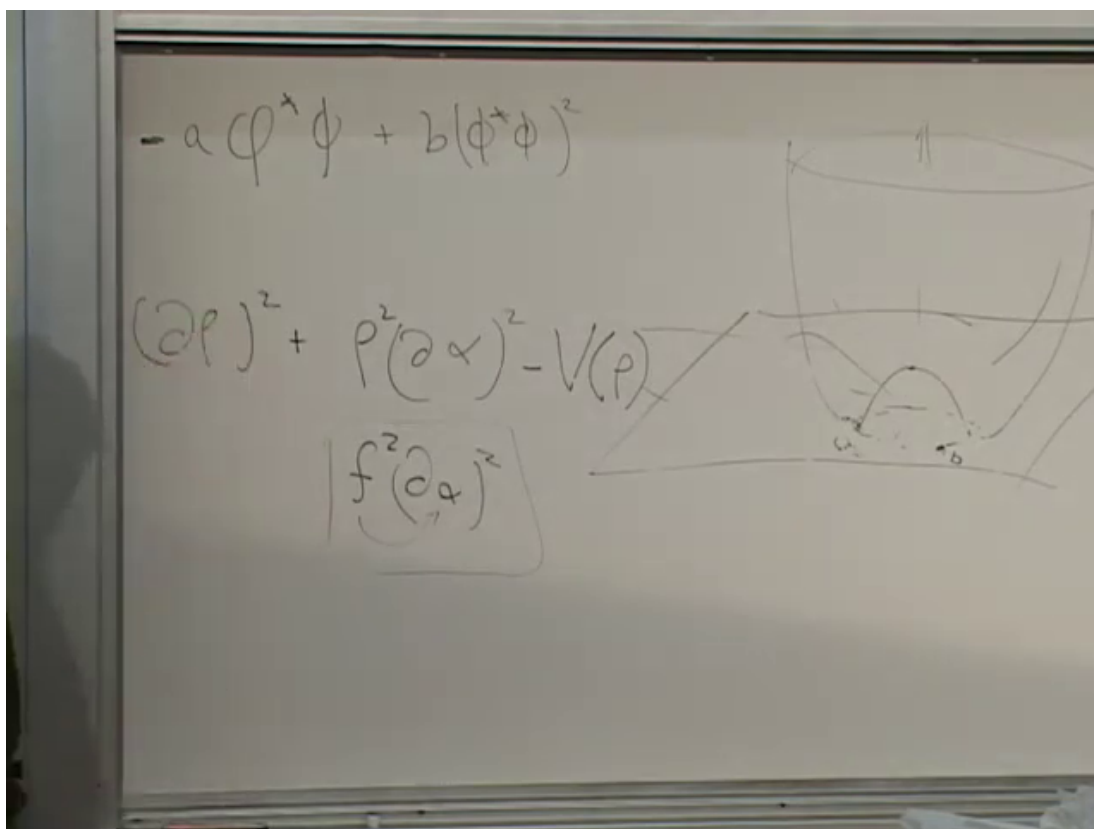


Figure 1.5: The polar-field Lagrangian, its boxed low-energy angular term, and the symmetry-breaking potential. Radial motion climbs the potential; angular motion follows the trough.

Lecture frame: 00:56:40.

Since $\phi^* \phi = \rho^2$,

$$\mathcal{L} = (\partial\rho)^2 + \rho^2(\partial\alpha)^2 - V(\rho^2). \quad (1.25)$$

The potential contains ρ but not α . That absence is the symmetry in its most useful form.

For a very low-energy excitation the field remains close to $\rho = f$. Then

$$\mathcal{L}_{\text{angular}} \simeq f^2(\partial_\mu\alpha)(\partial^\mu\alpha). \quad (1.26)$$

Define the canonically normalized angular field, up to the convention used for factors of 1/2, by

$$\beta = \sqrt{2} f \alpha, \quad \mathcal{L}_{\text{angular}} = \frac{1}{2}(\partial_\mu\beta)(\partial^\mu\beta). \quad (1.27)$$

There is no β^2 term. The quanta of this angular field are massless. They are the Goldstone bosons.

The radial field behaves differently. Displace ρ from f everywhere at once and it rolls back, overshoots, and oscillates about the trough. If $\rho = f + h$, expansion of Eq. (1.14) produces

$$V(f + h) = V(f) + \frac{1}{2}m_h^2 h^2 + \dots, \quad (1.28)$$

so h is massive. Geometrically, the angular direction is flat and the radial direction is curved.

Question from the audience. If the field is displaced radially, does it simply drop back down, and what happens next?^a

Response. With no angular motion it falls toward the minimum, overshoots, and oscillates radially. A homogeneous oscillation has zero spatial momentum but nonzero frequency; that frequency is the mass of the radial excitation. A homogeneous angular displacement, in contrast, lands on another vacuum and has no restoring force.

^aLecture timestamp: 00:57:10.

This is the content of the Goldstone principle in the present example: spontaneously breaking a continuous global symmetry produces a massless mode for each independent flat direction of the vacuum manifold. ²

1.6 Spin Waves, Strings, and Sound

The theorem is abstract, but its mechanism is familiar. In an isotropic ferromagnet all spins may be rotated together without changing the energy. Let the orientation vary very slowly across the sample and one obtains a spin wave. Its cost vanishes with its wave number, so it is the Goldstone mode of broken spin-rotation symmetry.

A long elastic string supplies another picture. Ignore gravity and mount the ends so that the whole string may be translated vertically. A uniform vertical translation costs no energy. Let the translation vary slowly along the string and it becomes a long-wavelength transverse wave. The uniform symmetry operation has been promoted to a gentle position-dependent deformation.

²Editorial clarification: The statement assumes the usual conditions of relativistic field theory. In nonrelativistic many-body systems the counting and dispersion of Nambu–Goldstone modes can be subtler, but the lecture’s long-wavelength conclusion remains the right one for this relativistic $U(1)$ model.

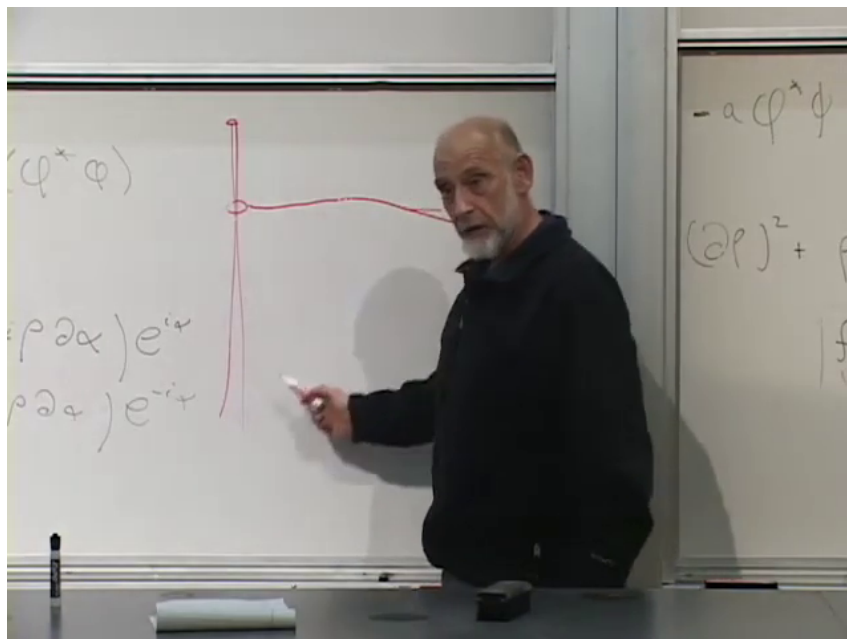


Figure 1.6: The slowly varying string profile drawn during the Goldstone-mode analogy. The physical point is the small slope of a long-wavelength deformation, not the vertical placement of the curve.

Lecture frame: 01:04:19.

Question from the audience. Can the string carry oscillation energy while having zero momentum?^a

Response. One must distinguish vertical mechanical momentum from wave momentum along the string. A standing wave can have zero net momentum because it is a superposition of modes with $+k$ and $-k$, while still carrying energy. In the field-theory analogy, momentum refers to the wave number along the axis; the vertical displacement stands for the internal field coordinate.

^aLecture timestamp: 01:04:41.

A slinky gives the longitudinal version. Translate the entire idealized infinite slinky along its axis and nothing changes. Translate neighboring parts by slightly different amounts and the result is a gentle compression wave. In a solid or fluid, long-wavelength sound similarly arises from collective displacements and density variations. Spin waves, transverse string waves, and sound are not the same microscopic excitation, but they share the decisive pattern: a uniform continuous operation costs no energy, and its slowly varying version becomes a soft mode.³

The terminology matters enough for a brief classroom joke: these are *Goldstone* bosons, named for Jeffrey Goldstone, not gauge bosons. The lecture notes that Goldstone was a friend of Susskind's and then pivots to the genuinely different subject of gauge invariance.

³Editorial clarification: In a crystal, acoustic phonons are conventionally described as Nambu–Goldstone modes of spontaneously broken continuous translations. The string and slinky are mechanical analogies used in the lecture to expose the same derivative-only energy structure.

1.7 From Global Phase Symmetry to Gauge Symmetry

The $U(1)$ phase symmetry has already appeared in the course as the symmetry associated with conservation of charge. Charged particle fields are complex, and multiplying such a field by a common phase leaves the global theory unchanged. The word electric may be used for intuition, though the charge could represent another $U(1)$ quantum number. The real photon is massless; the eventual target is a photon-like field such as the massive Z .

The new question is whether

$$\phi'(x) = e^{i\theta(x)}\phi(x) \quad (1.29)$$

can be a symmetry when θ varies arbitrarily through spacetime. The function $\theta(x)$ is a transformation parameter, not an additional dynamical field. We test the proposal by differentiating:

$$\partial_\mu \phi' = e^{i\theta} [\partial_\mu \phi + i(\partial_\mu \theta)\phi]. \quad (1.30)$$

The extra term was absent for constant θ . For the conjugate,

$$\partial_\mu \phi^{*'} = e^{-i\theta} [\partial_\mu \phi^* - i(\partial_\mu \theta)\phi^*]. \quad (1.31)$$

Their product is

$$\begin{aligned} \partial_\mu \phi^{*'} \partial^\mu \phi' &= \partial_\mu \phi^* \partial^\mu \phi i (\phi \partial_\mu \phi^* - \phi^* \partial_\mu \phi) \partial^\mu \theta \\ &\quad + \phi^* \phi \partial_\mu \theta \partial^\mu \theta. \end{aligned} \quad (1.32)$$

The potential remains invariant because $\phi^* \phi$ does, but the ordinary kinetic term fails. A global symmetry has not automatically become a local one.

Question from the audience. In the Mexican-hat picture, does this failure mean that there are bumps around the circular trough?^a

Response. No. The potential can remain perfectly flat in the angular direction. The added cost comes from gradients when the phase differs from one spacetime point to another. It is not a corrugation of the trough; it is the ordinary derivative term detecting spatial variation.

^aLecture timestamp: 01:19:37.

1.7.1 The compensating vector field

Nature's photon, W , Z , and gluon interactions are organized by local gauge symmetries, not merely by rigid transformations. To make the local phase operation a symmetry, enlarge the theory by a four-vector field A_μ . In electromagnetism A_0 is the scalar potential and the three spatial components form the vector potential. Their derivatives build the electric and magnetic fields.

Choose the internally consistent convention

$$\phi' = e^{i\theta} \phi, \quad A'_\mu = A_\mu - \partial_\mu \theta, \quad D_\mu = \partial_\mu + iA_\mu. \quad (1.33)$$

The board changes a sign while checking the cancellation; Eq. (1.33) fixes all three signs together. The gauge coupling and charge have temporarily been absorbed into A_μ .⁴

⁴Editorial clarification: With explicit charge q and coupling g , one may write $D_\mu = \partial_\mu + igqA_\mu$ and $A'_\mu = A_\mu - (gq)^{-1} \partial_\mu \theta$. The lecture suppresses these constants to keep the cancellation visible.

For the conjugate field,

$$D_\mu \phi^* = \partial_\mu \phi^* - i A_\mu \phi^*. \quad (1.34)$$

Now

$$\begin{aligned} D'_\mu \phi' &= (\partial_\mu + i A'_\mu) e^{i\theta} \phi \\ &= e^{i\theta} [\partial_\mu \phi + i(\partial_\mu \theta) \phi + i(A_\mu - \partial_\mu \theta) \phi] \\ &= e^{i\theta} D_\mu \phi. \end{aligned} \quad (1.35)$$

The unwanted derivative of θ is canceled by the transformation of A_μ . Likewise,

$$D'_\mu \phi^{*'} = e^{-i\theta} D_\mu \phi^*. \quad (1.36)$$

Their product is invariant. The adjective *covariant* means that the derivative transforms in the same simple way as the field itself.

1.8 Dynamics of the Gauge Field

Introducing A_μ only to repair a derivative would leave it without its own dynamics. The gauge-invariant field strength is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (1.37)$$

It is antisymmetric, so $F_{\mu\mu} = 0$. Its time–space components contain the electric field, while its space–space components contain the magnetic field. In familiar three-vector language the electromagnetic energy density is proportional to $\mathbf{E}^2 + \mathbf{B}^2$, whereas the Lorentzian Lagrangian is proportional to $\mathbf{E}^2 - \mathbf{B}^2$, with signs depending on the metric convention.

Question from the audience. What are the energy and Lagrangian of the electromagnetic field?^a

Response. The energy is built from $E^2 + B^2$; the Lagrangian uses $E^2 - B^2$. Covariantly the latter is written as a conventional multiple of $F_{\mu\nu} F^{\mu\nu}$. The antisymmetric tensor is the four-dimensional curl of the vector potential.

^aLecture timestamp: 01:33:20.

The field strength is itself gauge invariant:

$$\begin{aligned} F'_{\mu\nu} &= \partial_\mu (A_\nu - \partial_\nu \theta) - \partial_\nu (A_\mu - \partial_\mu \theta) \\ &= F_{\mu\nu} - \partial_\mu \partial_\nu \theta + \partial_\nu \partial_\mu \theta = F_{\mu\nu}. \end{aligned} \quad (1.38)$$

The final equality follows because ordinary mixed partial derivatives commute. A standard gauge-invariant Lagrangian is therefore

$$\boxed{\mathcal{L} = (D_\mu \phi)^* D^\mu \phi - V(\phi^* \phi) - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}} \quad (1.39)$$

up to the sign and normalization conventions chosen for the metric.

This construction promotes a global $U(1)$ symmetry to a local one, but it does so at the price—or, physically, with the gift—of a new vector field. That field is the electromagnetic potential in the Abelian example.

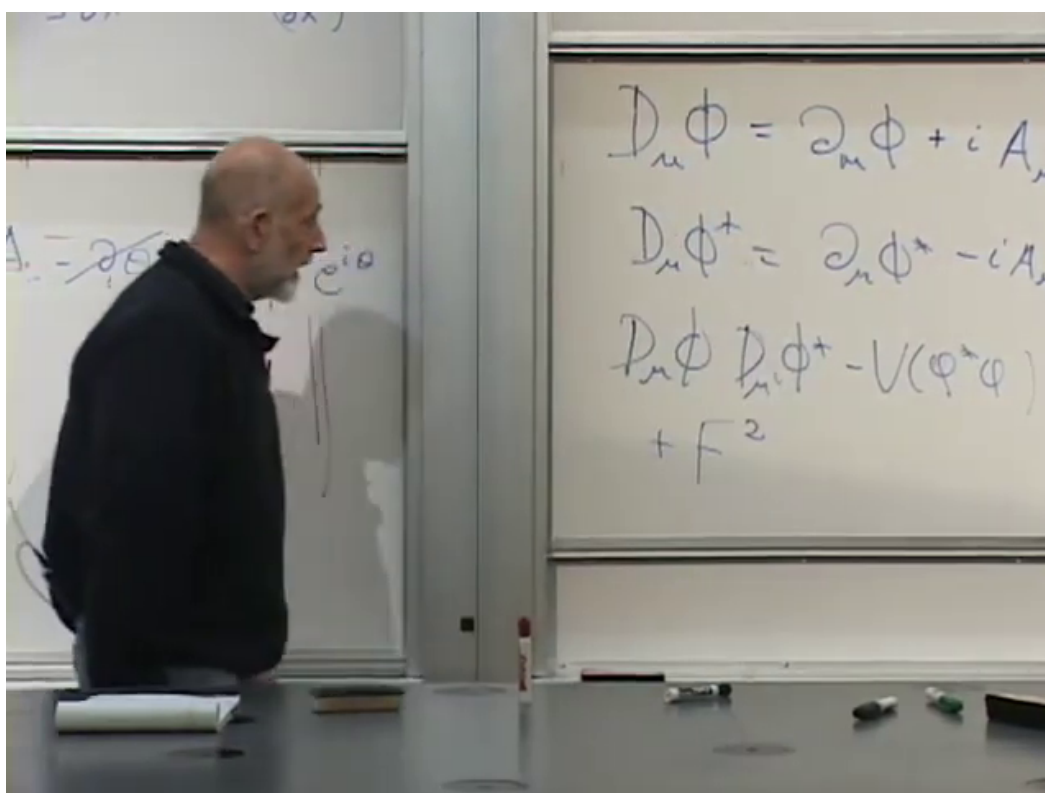


Figure 1.7: The covariant derivatives and the assembled gauge-invariant Lagrangian on the board. Numerical factors are suppressed there so that the transformation structure remains visible.

Lecture frame: 01:38:45.

Question from the audience. The Lagrangian we began with used ordinary derivatives. Why are we entitled to replace them by covariant derivatives and call the result the real theory?^a

Response. We are constructing a different, enlarged theory: charged matter plus a gauge field. Local gauge invariance determines their coupling. The resulting quantum theory is a coherent mathematical structure and agrees with how nature organizes the photon and weak gauge fields. Simply coupling a relativistic vector field inconsistently can introduce unphysical polarizations and violations of probability, so the gauge principle is not a cosmetic rewriting of the old free scalar theory.

^aLecture timestamp: 01:42:55.

Question from the audience. So adding the new field is the essential justification for the modification?^a

Response. Yes. Without A_μ , the derivative of a locally rotated field contains an uncanceled $\partial_\mu\theta$. With A_μ and its transformation law, the cancellation works. In geometric language A_μ is a connection on a bundle; that deeper mathematics is real and useful, but the algebra above contains the part needed here.

^aLecture timestamp: 01:44:57.

1.9 Why a Bare Gauge-Boson Mass Is Forbidden

For an ordinary bosonic field, a mass is represented by a quadratic term. It is tempting to add

$$\Delta\mathcal{L}_{\text{Proca}} = \frac{1}{2}m^2 A_\mu A^\mu. \quad (1.40)$$

But under $A_\mu \mapsto A_\mu - \partial_\mu\theta$,

$$A'_\mu A'^\mu = A_\mu A^\mu - 2A^\mu \partial_\mu\theta + \partial_\mu\theta \partial^\mu\theta, \quad (1.41)$$

so Eq. (1.40) is not gauge invariant. We cannot place a mass into this gauge theory by hand while retaining its local symmetry.

This is the apparent impasse. Gauge invariance explains the massless photon and protects the theory from unphysical degrees of freedom, but the W and Z are massive. The escape cannot be an explicit vector mass. It must arise from the fields already present in a gauge-invariant Lagrangian.

Question from the audience. Does introducing gauge invariance itself produce the Goldstone boson?^a

Response. No. The Goldstone boson arose when a continuous *global* symmetry was spontaneously broken, before the vector potential was introduced. Once the symmetry is local, a varying phase can be compensated by A_μ . In the broken gauge theory the angular mode is no longer an independent massless particle; it supplies the additional polarization of the now-massive vector field. The next lecture will derive that statement rather than merely announce it.

^aLecture timestamp: 01:45:51.

Question from the audience. Does any of this explain why rivers meander? ^a

Response. No direct connection is known here. Fluid eddies and even the shapes of organisms can certainly be described with symmetry methods, and one should never be astonished to find field-theory ideas in unexpected places, but the lecture claims no gauge theory of river meanders.

^aLecture timestamp: 01:47:03.

A photograph of a chalkboard with the equation $+ \frac{m^2}{2} A_\mu A^\mu$ written in blue chalk. The equation is centered on the board. There is a small mark in the bottom right corner of the board.

Figure 1.8: The proposed bare vector mass $\frac{1}{2}m^2 A_\mu A^\mu$. Its failure under $A_\mu \mapsto A_\mu - \partial_\mu \theta$ is the obstruction that sets up the Higgs mechanism.

Lecture frame: 01:39:48.

1.10 Where the Lecture Stops

We now have the two ingredients promised at the beginning. A continuous global symmetry with a circle of vacua has a flat angular direction, and its slow variation is a massless Goldstone field. A local symmetry requires a gauge connection and covariant derivatives, and it forbids the direct vector mass in Eq. (1.40).

The next step is to use the broken potential of Eq. (1.14) inside the gauge-invariant theory of Eq. (1.39). Then two things happen together: the separate Goldstone particle disappears from the physical spectrum, and the vector field acquires a mass. In the usual pedagogical language the gauge symmetry is spontaneously broken; more precisely, the gauge redundancy remains while the theory enters a Higgs phase with a massive vector excitation.⁵

That is the Higgs phenomenon. We have not yet performed it. This lecture has done the necessary harder work of making the problem precise enough that the solution, when it comes, will be unavoidable.

⁵Editorial clarification: A local gauge symmetry is a redundancy of description rather than an ordinary physical symmetry acting between distinct observable states. The lecture uses the standard shorthand of spontaneous gauge-symmetry breaking. The gauge-invariant physical statement is the rearrangement of degrees of freedom in the Higgs phase.